UNIT 1

EVOLUTION OF MACHINE LEARNING

Machine learning is a data analysis method that automates analytical model building. It is a branch of artificial intelligence based on the idea that systems can learn from data, identify patterns and make decisions with minimal human intervention. Because of new computing technologies, machine learning today is not like machine learning of the past. It was born from pattern recognition and the theory that computers can learn without being programmed to perform specific tasks; researchers interested in artificial intelligence wanted to see if computers could learn from data. The iterative aspect of machine learning is important because as models are exposed to new data, they are able to independently adapt. They learn from previous computations to produce reliable, repeatable decisions and results. It's a science that's not new – but one that has gained fresh momentum. While many machine learning algorithms have been around for a long time, the ability to automatically apply complex mathematical calculations to big data – over and over, faster and faster – is a recent development.

Here are a few widely publicized examples of machine learning applications you may be familiar with:

- The heavily hyped, self-driving Google car? The essence of machine learning.
- Online recommendation offers such as those from Amazon and Netflix? Machine learning applications for everyday life.
- Knowing what customers are saying about you on Twitter? Machine learning combined with linguistic rule creation.
- Fraud detection? One of the more obvious, important uses in our world today.

Why is machine learning important?

Resurging interest in machine learning is due to the same factors that have made data mining and Bayesian analysis more popular than ever. Things like growing volumes and varieties of available data, computational processing that is cheaper and more powerful, and affordable data storage. All of these things mean it's possible to quickly and automatically produce models that can analyze bigger, more complex data and deliver faster, more accurate results – even on a very large scale. And by building precise models, an organization has a better chance of identifying profitable opportunities – or avoiding unknown risks.

What's required to create good machine learning systems?

- Data preparation capabilities.

- Algorithms – basic and advanced.
- Automation and iterative processes.
- Scalability.
- Ensemble modeling.

Did you know?

- In machine learning, a target is called a label.
- In statistics, a target is called a dependent variable.
- A variable in statistics is called a feature in machine learning.
- A transformation in statistics is called feature creation in machine learning.

Who's using it?

Most industries working with large amounts of data have recognized the value of machine learning technology. By gleaning insights from this data – often in real time – organizations are able to work more efficiently or gain an advantage over competitors.

Financial services
Banks and other businesses in the financial industry use machine learning technology for two key purposes: to identify important insights in data and prevent fraud. The insights can identify investment opportunities, or help investors know when to trade. Data mining can also identify clients with high-risk profiles or use cyber surveillance to pinpoint warning signs of fraud.

Government
Government agencies such as public safety and utilities have a particular need for machine learning since they have multiple sources of data that can be mined for insights. Analyzing sensor data, for example, identifies ways to increase efficiency and save money. Machine learning can also help detect fraud and minimize identity theft.

Healthcare
Machine learning is a fast-growing trend in the healthcare industry, thanks to the advent of wearable devices and sensors that can use data to assess a patient's health in real time. The technology can also help medical experts analyze data to identify trends or red flags that may lead to improved diagnoses and treatment.

Marketing and sales
Websites recommending items you might like based on previous purchases are using machine learning to analyze your buying history – and promote other items you'd be interested in. The ability to capture data, analyze it and use it to personalize a shopping experience (or implement a marketing campaign) is the future of retail.

Oil and gas
Finding new energy sources. Analyzing minerals in the ground. Refinery sensor failure predicting. Streamlining oil distribution to make it more efficient and cost-effective. The number of machine learning use cases for this industry is vast – and still expanding.

Transportation
Analyzing data to identify patterns and trends is key to the transportation industry, which relies on making routes more efficient and predicting potential problems to increase profitability. The data analysis and modeling aspects of machine learning are important tools to delivery companies, public transportation and other transportation organizations.

What are some popular machine learning methods?

Two of the most widely adopted machine learning methods are supervised learning and unsupervised learning – but there are also other methods of machine learning. Here's an overview of the most popular types.
Supervised learning algorithms are trained using labeled examples, such as an input where the desired output is known. For example, a piece of equipment could have data points labeled either "F" (failed) or "R" (runs). The learning algorithm receives a set of inputs along with the corresponding correct outputs, and the algorithm learns by comparing its actual output with correct outputs to find errors. It then modifies the model accordingly. Through methods like classification, regression, prediction and gradient boosting, supervised learning uses patterns to predict the values of the label on additional unlabeled data. Supervised learning is commonly used in applications where historical data predicts likely future events. For example, it can anticipate when credit card transactions are likely to be fraudulent or which insurance customer is likely to file a claim.
Unsupervised learning is used against data that has no historical labels. The system is not told the "right answer." The algorithm must figure out what is being shown. The goal is to explore the data and find some structure within. Unsupervised learning works well on transactional data. For example, it can identify segments of customers with similar attributes who can then be treated similarly in marketing campaigns. Or it can find the main attributes that separate customer segments from each other. Popular techniques include self-organizing maps, nearest-neighbor mapping and singular value decomposition. These algorithms are also used to segment text topics, recommend items and identify data outliers.
Semisupervised learning is used for the same applications as supervised learning. But it uses both labeled and unlabeled data for training – typically a small amount of labeled data with a large amount of unlabeled data (because unlabeled data is

less expensive and takes less effort to acquire). This type of learning can be used with methods such as classification, regression and prediction. Semisupervised learning is useful when the cost associated with labeling is too high to allow for a fully labeled training process. Early examples of this include identifying a person's face on a webcam.

Reinforcement learning is often used for robotics, gaming and navigation. With reinforcement learning, the algorithm discovers through trial and error which actions yield the greatest rewards. This type of learning has three primary components: the agent (the learner or decision maker), the environment (everything the agent interacts with) and actions (what the agent can do). The objective is for the agent to choose actions that maximize the expected reward over a given amount of time. The agent will reach the goal much faster by following a good policy. So, the goal in reinforcement learning is to learn the best policy.

Humans can typically create one or two good models a week; machine learning can create thousands of models a week.

Exercises

1. Answer the following questions.

a) What is machine learning and what is it based on?
b) What was machine learning born from?
c) Why is the iterative aspect of machine learning important?
d) Give and describe some examples of machine learning applications.
e) Which factors resurged the interest in machine learning?
f) Name and explain some factors required to create good machine learning systems.
g) Give some terminological differences between statistics and machine learning.
h) What is the role of machine learning in financial services?
i) What is the role of machine learning in government agencies?
j) What is the role of machine learning in healthcare?
k) What is the role of machine learning in marketing and sales?
l) What is the role of machine learning relating to oil and gas?
m) What is the role of machine learning in transportation?
n) What do you know about supervised learning as a machine learning method?
o) What do you know about unsupervised learning as a machine learning method?
p) What do you know about semisupervised learning as a machine learning method?
q) What do you know about reinforcement learning as a machine learning method?

2. Explain/describe the following terms and expressions, on the basis of their use in the text. Use a dictionary or check online if necessary.

artificial intelligence	
pattern recognition	
fraud detection	
data mining	
iterative processes	
dependent variable	
cyber surveillance	
identity theft	
marketing campaign	
fraudulent	
historical labels	

3. Match the terms/expressions with their definitions.

Machine learning	implies the algorithm which discovers through trial and error which actions yield the greatest rewards.
Supervised learning	is a branch of artificial intelligence based on the idea that systems can learn from data, identify patterns and make decisions with minimal human intervention.
Unsupervised learning	is used for the same applications as supervised learning, but it uses both labeled and unlabeled data for training – typically a small amount of labeled data with a large amount of unlabeled data.
Semisupervised learning	algorithms are trained using labeled examples, such as an input where the desired output is known.
Reinforcement learning	is used against data that has no historical labels, so that the system is not told the "right answer" and the algorithm must figure out what is being shown.

4. Write the descriptions of the modified nouns from the table.

data analysis	method	
machine learning	algorithms	
data preparation	capabilities	
refinery sensor failure	predicting	
nearest-neighbor	mapping	
singular value	decomposition	

5. Complete the text with the given words and expressions.

optimize | experience | decision | data | pattern | data | error | prediction | accuracy | model | data | accuracy | prediction| artificial | algorithm

Machine learning (ML) is a branch of ___________ intelligence (AI) focused on enabling computers and machines to imitate the way that humans learn, to perform tasks autonomously, and to improve their performance and accuracy through ___________ and exposure to more ___________.
The learning system of a machine learning ___________ can be broken out into three main parts.
___________ Process: In general, machine learning algorithms are used to make a ___________ or classification. Based on some input ___________, which can be labeled or unlabeled, your algorithm will produce an estimate about a ___________ in the data.

___________ Function: An error function evaluates the ___________ of the model. If there are known examples, an error function can make a comparison to assess the ___________ of the model.

___________ Optimization Process: If the model can fit better to the ___________ points in the training set, then weights are adjusted to reduce the discrepancy between the known example and the model estimate. The algorithm will repeat this iterative "evaluate and ___________" process, updating weights autonomously until a threshold of ___________ has been met.

unstructured | algorithm | human | categories | data | neuron | threshold | layers| neural | vision | artificial intelligence | recognition | neural networks| supervised | structured

Since deep learning and machine learning tend to be used interchangeably, it's worth noting the nuances between the two. Machine learning, deep learning, and neural networks are all sub-fields of ___________. However, neural networks is actually a sub-field of machine learning, and deep learning is a sub-field of ___________.

The way in which deep learning and machine learning differ is in how each ___________ learns. "Deep" machine learning can use labeled datasets, also known as ___________ learning, to inform its algorithm, but it doesn't necessarily require a labeled dataset. The deep learning process can ingest ___________ data in its raw form (e.g., text or images), and it can automatically determine the set of features which distinguish different ___________ of data from one another. This eliminates some of the human intervention required and enables the use of large amounts of ___________. It is possible to think of deep learning as "scalable machine learning." Classical or "non-deep" machine learning is more dependent on ___________ intervention to learn. Human experts determine the set of features to understand the differences between data inputs, usually requiring more ___________ data to learn.

Neural networks, or artificial neural networks (ANNs), are comprised of node layers, containing an input layer, one or more hidden layers, and an output layer. Each node, or artificial ___________, connects to another and has an associated weight and threshold. If the output of any individual node is above the specified ___________ value, that node is activated, sending data to the next layer of the network. Otherwise, no data is passed along to the next layer of the network by that node. The "deep" in deep learning is just referring to the number of ___________ in a neural network. A neural network that consists of more than three layers—which would be inclusive of the input and the output—can be considered a deep

learning algorithm or a deep neural network. A neural network that only has three layers is just a basic ___________ network.
Deep learning and neural networks are credited with accelerating progress in areas such as computer ___________, natural language processing, and speech ___________.